

DUKI, Russian Federation

WMO: 314820

Lat: 51.717N

Lon: 135.933E

Elev: 433

StdP: 14.47

Time Zone: 10.00 (E10)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-37.9	-44.7	0.4	-40.8	-41.2	0.5	-37.6	19.7	7.6	16.9	10.6	0.1	N/A	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	86.5	70.0	83.0	68.6	79.4	66.7	73.1	82.0	70.7	79.7	68.5	77.1	3.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	112.1	79.1	67.5	102.7	75.9	65.3	94.9	73.5	37.1	82.1	34.9	79.9	33.1	78.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	10.8	8.6	-46.2	92.0	5.6	3.4	-50.2	94.4	-53.5	96.3	-56.6	98.2	-60.7	100.7
			-45.8	77.5	4.8	3.8	-49.3	80.2	-52.1	82.5	-54.8	84.6	-58.2	87.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	-16.3	-4.5	11.5	32.9	46.0	58.9	65.3	62.6	50.1	33.6	9.0	-10.9
	DBStd	29.65	10.90	11.26	10.45	7.88	7.87	7.71	6.11	5.41	6.42	8.68	10.92	11.02
	HDD50	9186	2057	1527	1192	514	178	15	0	0	78	508	1231	1886
	HDD65	13529	2522	1947	1657	963	590	212	74	113	447	972	1681	2351
	CDD50	1281	0	0	0	1	54	281	473	390	81	1	0	0
	CDD65	150	0	0	0	0	1	29	82	38	0	0	0	0
	CDH74	2178	0	0	0	0	79	597	1054	436	12	0	0	0
	CDH80	652	0	0	0	0	13	184	327	127	1	0	0	0
	WSAvg	2.3	1.3	1.9	2.6	3.3	3.4	2.5	2.2	2.1	2.3	2.7	2.1	1.2
	PrecAvg	25.4	0.7	0.6	0.9	1.7	3.0	3.2	4.4	4.3	2.8	2.0	1.1	1.0
(15) Precipitation	PrecMax	32.1	2.2	1.9	2.4	3.7	5.7	5.7	8.6	9.2	5.4	5.4	2.4	3.4
	PrecMin	15.3	0.1	0.2	0.1	0.6	0.9	0.9	0.0	1.5	0.7	0.6	0.2	0.1
	PrecStd	4.5	0.6	0.4	0.6	0.9	1.3	1.1	1.9	2.0	1.3	1.2	0.7	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.8	29.0	50.3	65.4	83.2	89.4	90.8	89.6	75.9	63.7	37.8	24.8
		MCWB	20.6	24.9	40.4	48.6	59.5	69.8	71.9	73.5	61.6	48.9	30.9	22.3
	2%	DB	17.8	23.4	42.0	58.2	76.4	85.1	87.0	83.5	71.2	58.1	32.9	19.0
		MCWB	16.5	21.1	33.7	44.5	57.8	68.0	70.9	69.9	60.1	46.3	28.7	16.5
	5%	DB	11.0	19.4	35.2	52.5	70.8	81.0	83.6	78.9	66.8	52.7	29.6	13.4
		MCWB	9.6	17.3	29.5	41.3	55.1	65.5	70.2	67.7	58.2	42.8	25.4	11.8
	10%	DB	2.5	14.7	30.8	47.5	65.0	76.6	80.3	74.6	62.2	48.1	26.0	8.6
		MCWB	1.3	12.6	26.2	38.1	51.8	63.3	68.8	65.4	55.2	40.3	23.0	7.5
	0.4%	WB	20.9	25.7	40.8	49.2	64.6	75.9	76.5	75.9	65.6	50.7	32.8	22.5
		MCDB	21.4	27.9	50.6	62.4	78.9	87.3	86.1	85.4	72.0	60.0	36.0	24.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	16.7	21.5	34.0	44.9	60.0	70.1	73.5	71.8	62.0	46.8	29.3	16.6
		MCDB	17.9	23.1	41.4	56.8	71.9	81.4	82.5	80.1	69.0	55.6	33.0	18.7
	5%	WB	9.5	17.2	30.4	41.7	56.2	67.4	71.5	69.3	59.2	43.9	25.7	11.7
		MCDB	11.1	19.5	35.4	51.8	68.8	77.8	81.0	76.3	64.2	51.6	28.8	13.2
	10%	WB	1.5	12.6	25.9	38.6	52.9	64.5	69.6	67.0	56.8	40.9	23.2	7.4
		MCDB	2.5	14.6	30.6	46.3	64.3	73.2	78.6	73.5	61.1	47.0	26.2	8.6
(35) Mean Daily Temperature Range	5% DB	MDBR	27.6	30.2	30.0	21.5	23.1	25.3	22.3	19.2	20.0	20.7	22.7	23.5
		MCDBR	24.5	29.0	31.4	31.6	37.0	34.4	31.3	29.5	29.1	29.8	22.8	22.4
		MCWBR	22.6	25.5	23.7	18.8	20.1	18.5	17.5	17.3	18.0	17.8	18.0	20.6
		MCDBR	23.8	26.5	28.8	30.2	34.5	29.8	27.5	25.7	23.9	28.3	20.4	21.6
	5% WB	MCWBR	22.0	23.5	21.2	18.5	20.3	18.0	17.0	17.1	16.2	17.5	16.3	20.0
(40) Clear-Sky Solar Irradiance	taub		0.265	0.298	0.369	0.480	0.562	0.503	0.502	0.424	0.377	0.382	0.314	0.265
	taud		2.115	2.054	1.897	1.790	1.757	2.012	2.064	2.301	2.358	2.135	2.087	2.085
	Ebn at Noon		231	254	253	237	224	242	239	255	255	219	203	206
	Edh at Noon		23	34	49	62	67	53	49	37	31	32	24	21
(44) All-Sky Solar Radiation	RadAvg		469	848	1360	1652	1571	1719	1569	1371	1074	677	447	357
	RadStd		16	27	56	123	163	185	158	136	96	55	34	21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page